Mathematical Methods and Proofs

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Course / Assignment

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$$\dot{x} = \sigma(y - x),$$
$$\dot{y} = x(\rho - z) - y,$$
$$\dot{z} = xy - \beta z$$

σ=10, ρ=28, β=$\frac{8}{3}$

1 Problem set and notation

This starter demonstrates a proof-oriented assignment. Vectors are bold, $\mathbf{x} \in \mathbb{R}^n$; use $\|\mathbf{x}\|_2$ for fixed delimiters and $\left\|\frac{\mathbf{x}}{2}\right\|_2$ for automatic sizing.

Problem 1 (Linear algebra)**.** For the matrix below, compute its determinant and solve $A\mathbf{x} = \mathbf{b}$.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}. \tag{1}$$

Solution. The determinant is $2 \cdot 3 - 1 = 5$, and elimination gives $\mathbf{x} = (1/5, 3/5)^{\mathsf{T}}$. Substitution into equation (1) verifies both equations. □

Theorem 1.1 (Uniqueness)**.** *If $A \in \mathbb{R}^{n\times n}$ has $\det A \neq 0$, then $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $\mathbf{b} \in \mathbb{R}^n$.*

Proof. The matrix is invertible, so multiplication by A^{-1} gives $\mathbf{x} = A^{-1}\mathbf{b}$. Any two solutions have zero difference after multiplication by A, hence are equal. □

2 Calculus and differential equations

$$\int_0^1 x^2 \, \mathrm{d}x = \frac{1}{3}, \tag{2}$$

$$y'(t) + y(t) = 0, \quad y(0) = 1, \qquad y(t) = e^{-t}. \tag{3}$$

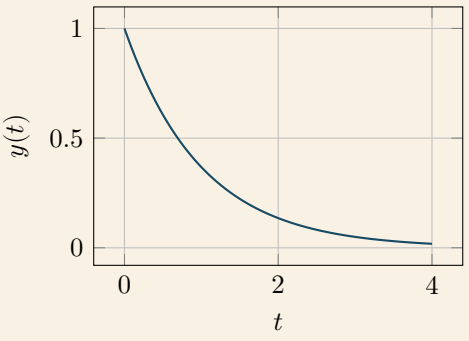


Figure 1: Analytical solution of equation (3); this is a function plot, not measured data.

3 Discrete mathematics and reporting

Definition 3.1. A graph is bipartite if its vertices admit a partition into two sets such that every edge joins vertices from different sets.

Problem 2 (Induction)**.** Prove that $\sum_{k=1}^{n} k = n(n+1)/2$ for every $n \in \mathbb{N}$, $n \geq 1$.

Solution. The case $n = 1$ holds. If the formula holds at n, then

$$\sum_{k=1}^{n+1} k = \frac{n(n+1)}{2} + n + 1 = \frac{(n+1)(n+2)}{2},$$

which completes the induction. □

Table 1: Exact quantities for the worked examples.

Quantity	Value	Verification
Determinant	5	Direct expansion
Solution	$(1/5, 3/5)^{\mathsf{T}}$	Substitution
Integral	$1/3$	Antiderivative

Symbolic computation. A computer algebra system may assist a derivation, but this document does not execute symbolic code. Record the input, tool version, assumptions, and independent check if you add computational results. Load the optional code module to include a transcript, or the bibliography module for cited sources.